MATHEMATICS PROJECT WORK RECORD :: FA-4

Topic: Mensuration

1. PRELIMINARY INFORMATION

Name of the Student: .....

Class & Section: .....

Roll Number: .....

Subject: Mathematics

Topic: Mensuration

2. AIM

To analyze a composite solid formed by a cylinder surmounted by a hemisphere, compute its total surface area and internal capacity step by step, and verify individual surface and volumetric additions.

3. OBJECTIVES

- Identify parameters: common base radius (r) and cylindrical height (h).
- Compute Curved Surface Area of cylinder ($2\pi rh$) and hemisphere ($2\pi r^2$).
- Calculate flat circular base area (πr^2) and sum them to obtain Total Surface Area.
- Determine composite volume by adding cylindrical volume ($\pi r^2 h$) and hemispherical volume ($(2/3)\pi r^3$).

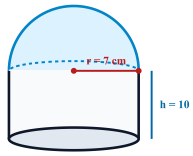
4. MATERIALS REQUIRED

Geometry instruments, millimeter scale, pen, pencil, eraser, and Class 10 Mathematics textbook.

5. PROCEDURE

1. Constructed a model of a cylinder surmounted by a hemisphere with r = 7 cm and h = 10 cm.
2. Calculated curved surface area of cylinder using $CSA_{cyl} = 2\pi rh$.
3. Calculated curved surface area of hemisphere using $CSA_{hemi} = 2\pi r^2$.
4. Added base circle area (πr^2) to find Total Surface Area: $TSA = CSA_{cyl} + CSA_{hemi} + Area_{base}$.
5. Calculated cylindrical volume $V_{cyl} = \pi r^2 h$ and dome volume $V_{hemi} = (2/3)\pi r^3$.
6. Summed both volumetric components to establish total solid capacity.

6. OBSERVATIONS AND STEP-BY-STEP CALCULATIONS



Given Parameters:

- Common Radius (r) = 7 cm
- Height of Cylinder (h) = 10 cm
- Value of $\pi = 22/7$
- **Base Circle Area:** $\pi r^2 = (22/7) \times 7 \times 7 = 22 \times 7 = \mathbf{154\ cm^2}$

I. SURFACE AREA CALCULATIONS

1. Curved Surface Area of Cylinder:

$CSA_{cyl} = 2\pi rh$

$= 2 \times (22/7) \times 7 \times 10$

$= 2 \times 22 \times 10$

$= 44 \times 10 = \mathbf{440\ cm^2}$

2. Curved Surface Area of Hemisphere:

$CSA_{hemi} = 2\pi r^2$

$= 2 \times (22/7) \times 7 \times 7$

$= 2 \times 22 \times 7$

$= 44 \times 7 = \mathbf{308\ cm^2}$

Total Surface Area (TSA):

$= CSA_{cyl} + CSA_{hemi} + Area_{base}$

$= 440 + 308 + 154 = \mathbf{902\ cm^2}$

II. VOLUME CALCULATIONS

1. Volume of Cylindrical Portion:

$V_{cyl} = \pi r^2 h$

$= (22/7) \times 7 \times 7 \times 10$

$= 22 \times 7 \times 10$

$= 154 \times 10 = \mathbf{1{,}540\ cm^3}$

2. Volume of Hemispherical Dome:

$V_{hemi} = (2/3)\pi r^3$

$= (2/3) \times (22/7) \times 7 \times 7 \times 7$

$= (2/3) \times 22 \times 49$

$= 2{,}156 / 3 = \mathbf{718.67\ cm^3}$

Total Capacity / Volume (V):

$= V_{cyl} + V_{hemi}$

$= 1{,}540 + 718.67 = \mathbf{2{,}258.67\ cm^3}$

7. CONCLUSION

- The total external surface area of the composite solid is **902 cm²**.
- The total volume (capacity) of the solid is **2,258.67 cm³**.

8. ACKNOWLEDGEMENT

I express my sincere thanks to my **Mathematics Teacher** for continuous guidance and support during this project.

9. RESOURCES

Telangana SCERT Class 10 Mathematics Textbook (Topic: *Mensuration*).

Signature of Student	**Signature of Mathematics Teacher**
Date: ____/____/2026	Verified